

Phu Gia Pham

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EDUCATION

University of Texas at San Antonio

Bachelor of Science in Computer Science, Minor in Business

San Antonio, TX

Aug. 2023 – Dec. 2026

San Antonio College

Associate of Science in Computer Science

San Antonio, TX

Aug. 2021 – May 2023

EXPERIENCE

San Antonio College

Professional Tutor (Mathematics & Computer Science)

San Antonio, TX

Aug. 2023 – Present

- Formulate adaptive pedagogical strategies to decompose complex mathematical proofs, advanced data structures, and algorithmic logic for 40+ undergraduate STEM students weekly.
- Design structured code-review frameworks and diagnostic debugging exercises, resulting in a measurable 15% average improvement in student conceptual retention and exam performance.
- Provide academic guidance and strategic mentorship, successfully driving student retention and critical problem-solving skills across multi-level Java, C++, and Python coursework.

University of Texas at San Antonio

Undergraduate Researcher

San Antonio, TX

Aug. 2025 – Present

- Conducted demographic bias mitigation research on Azure OpenAI's clinical recommendation models by architecting end-to-end evaluation pipelines utilizing large-scale, real-world healthcare datasets.
- Utilized Plotly to design and present complex disparate impact analyses, successfully identifying, quantifying, and documenting systemic bias gaps of up to 72% across critical target demographic subgroups.
- Presented actionable data-driven insights to academic stakeholders, proposing algorithmic guardrails to minimize downstream variance in automated medical decision systems.

San Antonio College

Student Ambassador & Peer Assistant

San Antonio, TX

Aug. 2021 – Aug. 2023

- Served as the primary liaison and peer coordinator for incoming STEM students, managing campus engagement initiatives and cross-departmental communication workflows.
- Facilitated technical orientation workshops and resource navigation, optimizing student onboarding pipelines for an active community of over 2,500 students.

PROJECTS

ML - Clinical AI Bias Mitigation | *Python, PyTorch, Azure OpenAI, SciKit-Learn*

Jan. 2026 – Present

- Engineered an automated NLP evaluation pipeline in Python to isolate and quantify demographic bias in LLM-generated medical treatment recommendations using counterfactual patient cohorts.
- Integrated Azure OpenAI APIs to dynamically query, parse, and extract clinical entities while rigorously validating generative model stability across 50+ structured prompt templates.
- Developed a Monte Carlo subsampling simulation to mathematically prove the statistical significance ($p < 0.05$) of latent demographic biases across age, race, and ethnicity variables.

Audio-Kinematic AI Pipeline | *Python, PyTorch, Librosa, Pandas*

Jan. 2026 – May 2026

- Developed an end-to-end data engineering pipeline capable of high-throughput ingestion, structural processing, and analytical preparation on massive robotic telemetry datasets exceeding 80,000 records.
- Implemented Multiple Sequence Alignment (MSA) logic to synchronize asynchronous sensor streams, ensuring precise spatial alignment across motor positions, velocity curves, and spatial paths.
- Demonstrated practical applications of cross-modal generation by translating raw acoustic features into highly predictable mechanical coordinate trajectories.

AI ChatBot System | *Python, PyTorch, Java, Discord API, Microservices*

July 2025 – Present

- Designed and deployed a modular conversational AI architecture utilizing localized large language model instances for high-throughput, real-time natural language understanding.

- Fine-tuned transformer models on customized conversational datasets to reinforce targeted emotional nuance, contextual memory tracking, and distinct persona traits.
- Containerized operational execution runtimes within automated worker microservices natively integrated into private production environments, handling 1,200+ daily interactions.

Food Classification System | *Python, PyTorch, Jupyter, Docker, Hugging Face* Mar. 2025 – May 2025

- Developed a custom CNN-based image classification network trained on the Food101 dataset, securing a 95% validation accuracy across highly complex target visual features.
- Designed robust data augmentation and normalization pipelines to mitigate overfitting risks caused by variable camera lighting, sizing distributions, and image resolution.
- Containerized the PyTorch inference environment using Docker and deployed the runtime directly to Hugging Face Spaces for public web API access.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL (PostgreSQL), JavaScript

Frameworks: React, Next.js, Node.js, Flask, JUnit, Tokenizers, FastAPI

Developer Tools: Git, Docker, Jupyter, Google Cloud Platform (GCP), VS Code, Hugging Face, OpenAI API

Libraries: Pandas, NumPy, Matplotlib, PyTorch, Torchvision, TorchText, Librosa, SciKit-Learn